ALGORITHM 1: High-Frequency Baseline Discrete Trading Agent

(policy-gradient update; PPO is one option)

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Input  : Historical prices {P_t}, fee c, epochs E, days-per-epoch K

Output : Optimised policy parameters θ*


Initialise policy θ and value φ randomly
FOR epoch = 1 to E DO
    FOR each trading day (episode) k = 1 to K DO
        Initialise state  s_0 = [ΔP_0, h_0 = 0]ᵀ
        Initialise buffer B = {}
        FOR minute t = 0 to T−1  (T = 390) DO
            Compute π_θ(a_t | s_t)
            Sample a_t ~ π_θ   where a_t ∈ {0:Hold, 1:Buy, 2:Sell}
            IF a_t = 1 THEN h_{t+1} = 1     (Buy / Long)
            IF a_t = 2 THEN h_{t+1} = 0     (Sell / Cash)
            IF a_t = 0 THEN h_{t+1} = h_t   (Hold)
            Observe  ΔP_{t+1} = (P_{t+1} − P_t) / P_t
            Reward   r_t = (h_t · ΔP_{t+1}) − c · |h_{t+1} − h_t|
            Next state s_{t+1} = [ΔP_{t+1}, h_{t+1}]ᵀ
            Store (s_t, a_t, r_t, s_{t+1}) in B
        END FOR
    END FOR
    Update θ by policy gradient / PPO loss on B
    Update value network φ
END FOR
RETURN θ*

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Iteration 1 only — 3 actions, binary holding, one stock, one day = one trajectory